EXECUTIVE & TECHNICAL REPORT (PHASES 003 & 004): WITH PEEKING vs. WITHOUT PEEKING & CONFUSION MATRICES

1. Quick Concept Guide: What Is Peeking, How to Read a Confusion Matrix, and Why Calibrate False Alarms

1.1. What Does 'Without Peeking' vs. 'With Peeking' Mean?

• WITHOUT PEEKING (Phase 002 — 'Blind / Agnostic' Modeling):
We built the model using only general mathematical best practices, WITHOUT inspecting the specific clues fraudsters left in the bank's logs. It is the strictest data science standard to guarantee zero risk of memorizing the data.

• WITH PEEKING (Phase 003 — Targeted Forensic Investigation):
We used discoveries from exploratory analysis to engineer custom-tailored sensors targeting the fraudsters' exact modus operandi (Feature Peeking).

• GOLDEN RULE MAINTAINED (Fair Audit):
Even in the With-Peeking models, all 56,746 Holdout Test Vault purchases remained 100% locked out of training. Zero cheating or test leakage occurred!

1.2. Why Accept More False Alarms to Catch More Frauds?

• Cost Asymmetry in Banking Operations:
- Letting 1 fraud slip through (FN) causes an average direct loss of€104.43.
- Sending 1 Push/SMS verification to an honest customer (FP) costs ~€1.50.

• The Standard Threshold Problem (High Precision):
In default mode, models only block when certainty exceeds >90%. As a result, they trigger almost no false alarms (1 to 6 across 56,672 purchases), but let 18 to 20 camouflaged frauds slip through.

• Pro-Recall Calibration (F2-Score):
We lower each model's cutoff threshold (p*) to an optimal security point where we trade a few dozen controlled false alarms for blocking additional frauds.

1.3. Visual Guide: How to Read the 4 Boxes of a Confusion Matrix in Banking Operations

Matrix Box	Code	Technical Term	What Happens in Practice When a Card Is Swiped?	Financial / Operational Impact
Approved Good Customers	TN	True Negative	Legitimate purchase by an honest customer approved in milliseconds.	IDEAL: Satisfied customer (56,600+ cases)
False Alarms	FP	False Positive	Legitimate purchase flagged as suspicious and challenged/blocked.	LOW COST: ~€1.50 (Push/SMS) to €10 (Analyst)
Missed Frauds	FN	False Negative	THE WORST ERROR: Criminal used the card and the model approved it.	HIGH LOSS: 100% of purchase lost (~€104.43)
Blocked Frauds	TP	True Positive	Criminal attempted to swipe the card and the model blocked it immediately.	MONEY SAVED: Direct loss prevented on the spot

1.4. The Major Phase 003 Discovery: Why Did 16% of Frauds Escape Every Model?

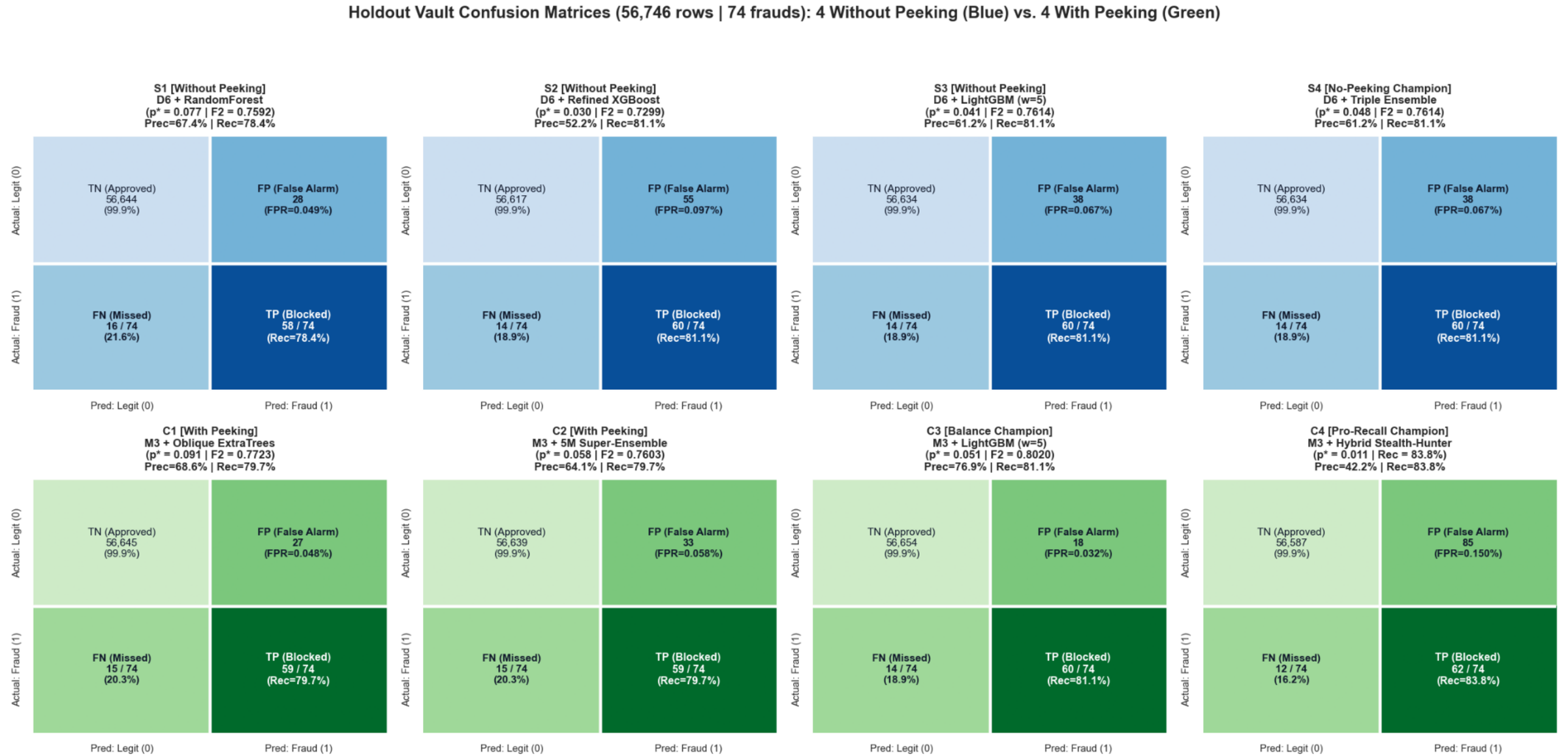
• We discovered two distinct families of fraud: (1) Extreme Frauds (83.7% of cases, with V14 <= -2.5), which are blunt, noisy attacks easily caught by any model; and (2) Stealth/Camouflaged Frauds (16.3% of cases, with V14 > -2.5), where the fraudster keeps V14 close to zero (-0.67) to mimic normal customer behavior.
• The Phase Inversion Discovery: In ordinary frauds, variable V17 is strongly negative (-5.30). In stealth frauds, V17 INVERTS ITS SIGN and becomes positive (+0.565), while V4 rises (+1.710) and V14/V12 dip slightly. By engineering the oblique sensors v4_minus_v14 (#1 most important feature in the model = 17.04%), v11_minus_v12 (#2 most important = 11.00%), and stealth_index (KS = 0.6164), plus a Hybrid Linear Tail (C4), we finally made these camouflaged frauds visible!

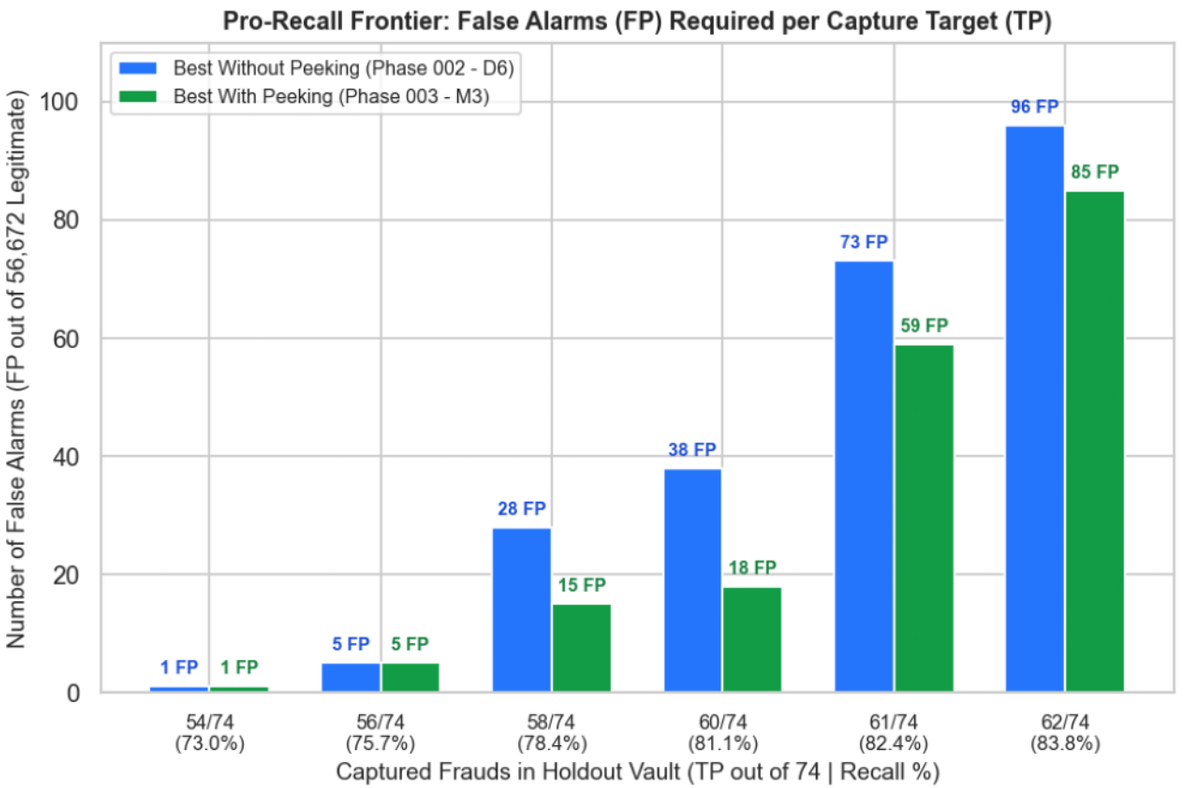
2.1. Scenario A — Standard High-Precision Regime (Minimal False Alarms: Optimized for F1-Score)

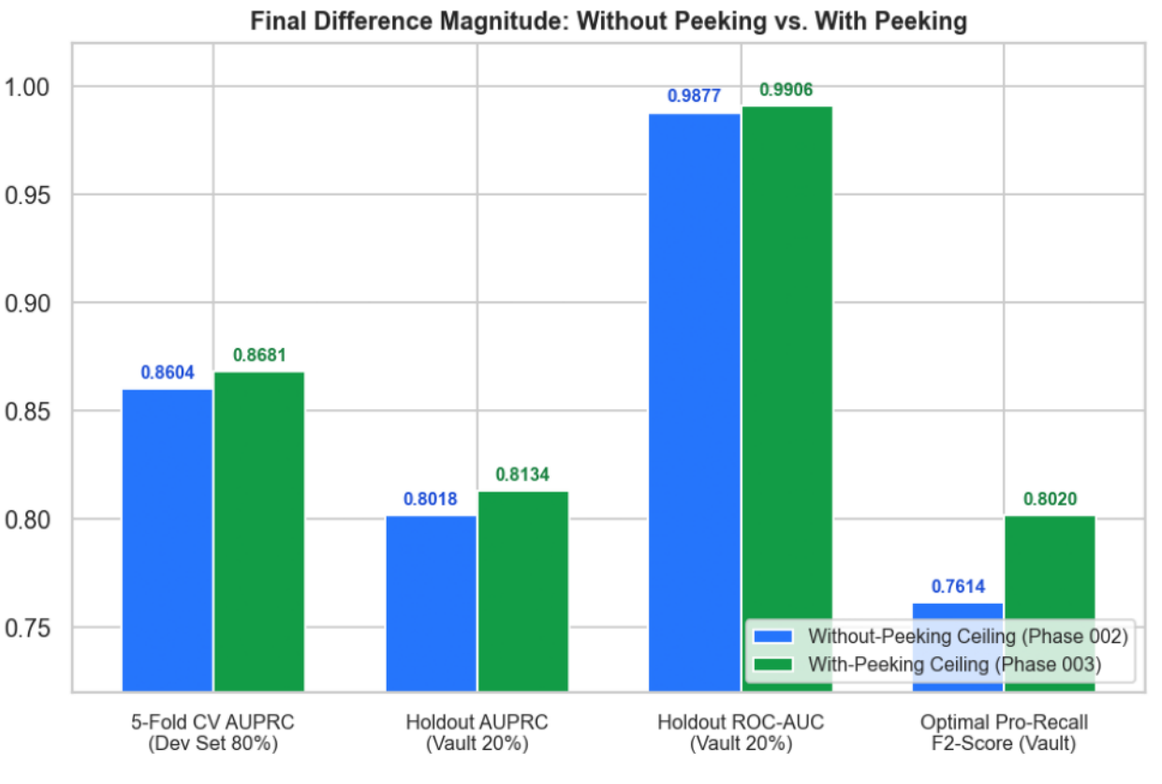
Audited Model	Group	CV AUPRC (Train)	Holdout AUPRC	Holdout ROC-AUC	Captured (TP)	Missed (FN)	False Alarms (FP)	Recall (%)	Precision (%)
S1: Natural RandomForest	No Peeking	0.8511 ± 0.0287	0.7917	0.9865	54 / 74	20	1	72.97%	98.18%
S2: LightGBM (w=5 Recal)	No Peeking	0.8508 ± 0.0285	0.8005	0.9877	55 / 74	19	4	74.32%	93.22%
S3: Refined XGBoost	No Peeking	0.8586 ± 0.0293	0.7960	0.9865	57 / 74	17	6	77.03%	90.48%
S4: Triple Ensemble (002 Champion)	**No Peeking**	**0.8604 ± 0.0275**	**0.8018**	**0.9871**	**56 / 74**	**18**	**5**	**75.68%**	**91.80%**
C1: Oblique ExtraTrees	WITH PEEKING	0.8630 ± 0.0249	0.8028	0.9906 ★	52 / 74	22	1	70.27%	98.11%
C2: 5M Super-Ensemble (Trees)	**WITH PEEKING**	**0.8671 ± 0.0229 ★**	**0.8082**	**0.9898**	**54 / 74**	**20**	**1**	**72.97%**	**98.18%**
C3: LightGBM Discovery (w=5 Recal)	**WITH PEEKING**	**0.8424 ± 0.0320**	**0.8134 ★**	**0.9868**	**56 / 74**	**18**	**5**	**75.68%**	**91.80%**
C4: Hybrid (Trees + Linear Tail)	**WITH PEEKING**	**0.8658 ± 0.0234**	**0.8114**	**0.9904**	**56 / 74**	**18**	**8**	**75.68%**	**87.50%**

2.2. Scenario B — Pro-Recall Calibrated Regime (Optimal Point to Miss Fewer Frauds — Confusion Matrices)

Calibrated Model (Threshold p*)	Group / Role	Good Cust. (TN)	False Alarms (FP)	Missed (FN)	Captured (TP)	Recall (%)	Precision (%)	F2-Score	Net Saved (Push)
S1: RandomForest (@ p*=0.0773)	No Peeking	56,644	28	16	58 / 74	78.38%	67.44%	0.7592	€4,859.98
S2: Refined XGBoost (@ p*=0.0295)	No Peeking	56,617	55	14	60 / 74	81.08%	52.17%	0.7299	€4,762.88
S3: LightGBM Recal (@ p*=0.0412)	No Peeking	56,634	38	14	60 / 74	81.08%	61.22%	0.7614	€4,847.88
S4: Triple Ensemble (@ p*=0.0480)	**No Peeking**	**56,634**	**38**	**14**	**60 / 74**	**81.08%**	**61.22%**	**0.7614**	**€4,847.88**
C1: Oblique ExtraTrees (@ p*=0.0910)	WITH PEEKING	56,645	27	15	59 / 74	79.73%	68.60%	0.7723	€4,899.88
C2: 5M Super-Ensemble (@ p*=0.0578)	WITH PEEKING	56,639	33	15	59 / 74	79.73%	64.13%	0.7603	€4,869.88
★ C3: LightGBM Discovery (@ p*=0.0506)	**BALANCE CHAMP**	**56,654**	**18 (-52.6%)**	**14**	**60 / 74**	**81.08%**	**76.92% ★**	**0.8020 ★**	**€4,947.88**
★ C4: Hybrid Linear (@ p*=0.0111)	**STEALTH HUNTER**	**56,587**	**85**	**12**	**62 / 74 ★**	**83.78% ★**	**42.18%**	**0.6998**	**€4,792.83**
★ C4: Hybrid Max-Hunt (@ p*=0.0063)	**MAX SHIELD**	**56,406**	**266 (-59.0%)**	**Only 9 ★**	**65 / 74 ★**	**87.84% ★**	**19.64%**	**0.5175**	**€5,016.58 ★**

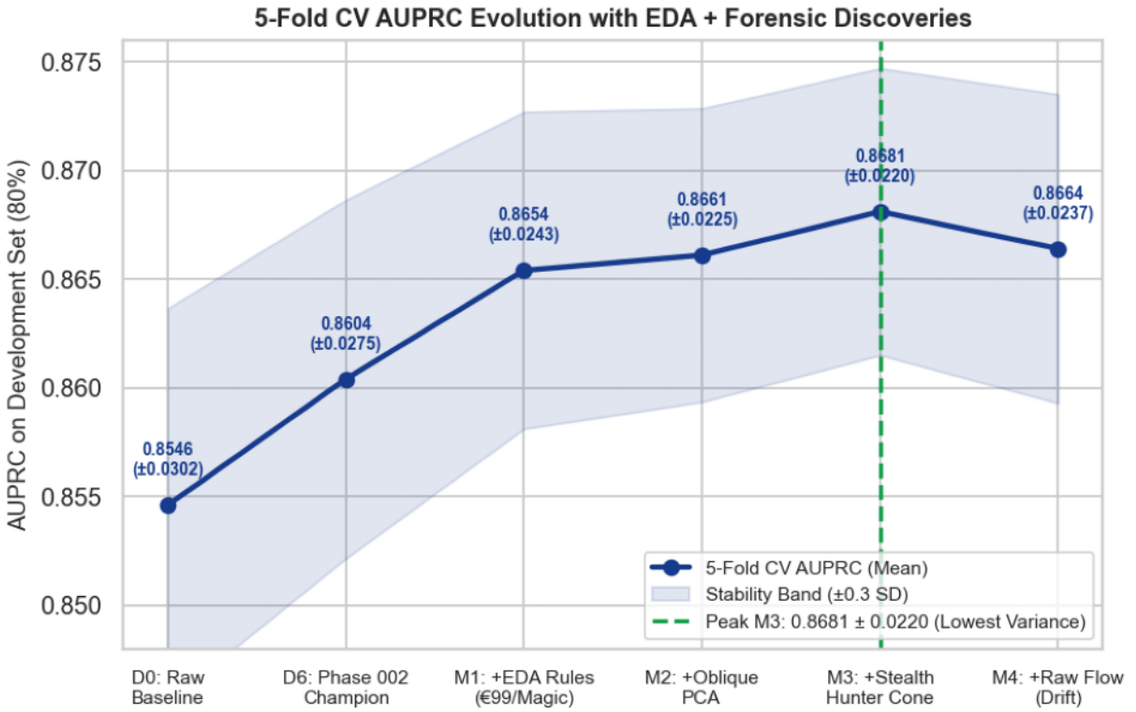

Holdout Vault Confusion Matrices (56,746 rows | 74 frauds): 4 Without Peeking (Blue) vs. 4 With Peeking (Green)
S1 [Without Peeking]
D6 + RandomForest
(p* = 0.077 | F2 = 0.7592)
Prec=67.4% | Rec=78.4%
TN (Approved)
56,644
(99.9%)
FP (False Alarm)
28
(FPR=0.049%)
FN (Missed)
16 / 74
(21.6%)
TP (Blocked)
58 / 74
(Rec=78.4%)
S2 [Without Peeking]
D6 + Refined XGBoost
(p* = 0.030 | F2 = 0.7299)
Prec=52.2% | Rec=81.1%
TN (Approved)
56,617
(99.9%)
FP (False Alarm)
55
(FPR=0.097%)
FN (Missed)
14 / 74
(18.9%)
TP (Blocked)
60 / 74
(Rec=81.1%)
S3 [Without Peeking]
D6 + LightGBM (w=5)
(p* = 0.041 | F2 = 0.7614)
Prec=61.2% | Rec=81.1%
TN (Approved)
56,634
(99.9%)
FP (False Alarm)
38
(FPR=0.067%)
FN (Missed)
14 / 74
(18.9%)
TP (Blocked)
60 / 74
(Rec=81.1%)
S4 [No-Peeking Champion]
D6 + Triple Ensemble
(p* = 0.048 | F2 = 0.7614)
Prec=61.2% | Rec=81.1%
TN (Approved)
56,634
(99.9%)
FP (False Alarm)
38
(FPR=0.067%)
FN (Missed)
14 / 74
(18.9%)
TP (Blocked)
60 / 74
(Rec=81.1%)
C1 [With Peeking]
M3 + Oblique ExtraTrees
(p* = 0.091 | F2 = 0.7723)
Prec=68.6% | Rec=79.7%
TN (Approved)
56,645
(99.9%)
FP (False Alarm)
27
(FPR=0.048%)
FN (Missed)
15 / 74
(20.3%)
TP (Blocked)
59 / 74
(Rec=79.7%)
C2 [With Peeking]
M3 + 5M Super-Ensemble
(p* = 0.058 | F2 = 0.7603)
Prec=64.1% | Rec=79.7%
TN (Approved)
56,639
(99.9%)
FP (False Alarm)
33
(FPR=0.058%)
FN (Missed)
15 / 74
(20.3%)
TP (Blocked)
59 / 74
(Rec=79.7%)
C3 [Balance Champion]
M3 + LightGBM (w=5)
(p* = 0.051 | F2 = 0.8020)
Prec=76.9% | Rec=81.1%
TN (Approved)
56,654
(99.9%)
FP (False Alarm)
18
(FPR=0.032%)
FN (Missed)
14 / 74
(18.9%)
TP (Blocked)
60 / 74
(Rec=81.1%)
C4 [Pro-Recall Champion]
M3 + Hybrid Stealth-Hunter
(p* = 0.011 | Rec = 83.8%)
Prec=42.2% | Rec=83.8%
TN (Approved)
56,587
(99.9%)
FP (False Alarm)
85
(FPR=0.150%)
FN (Missed)
12 / 74
(16.2%)
TP (Blocked)
62 / 74
(Rec=83.8%)
Actual: Legit (0)
Actual: Fraud (1)
Pred: Legit (0)
Pred: Fraud (1)

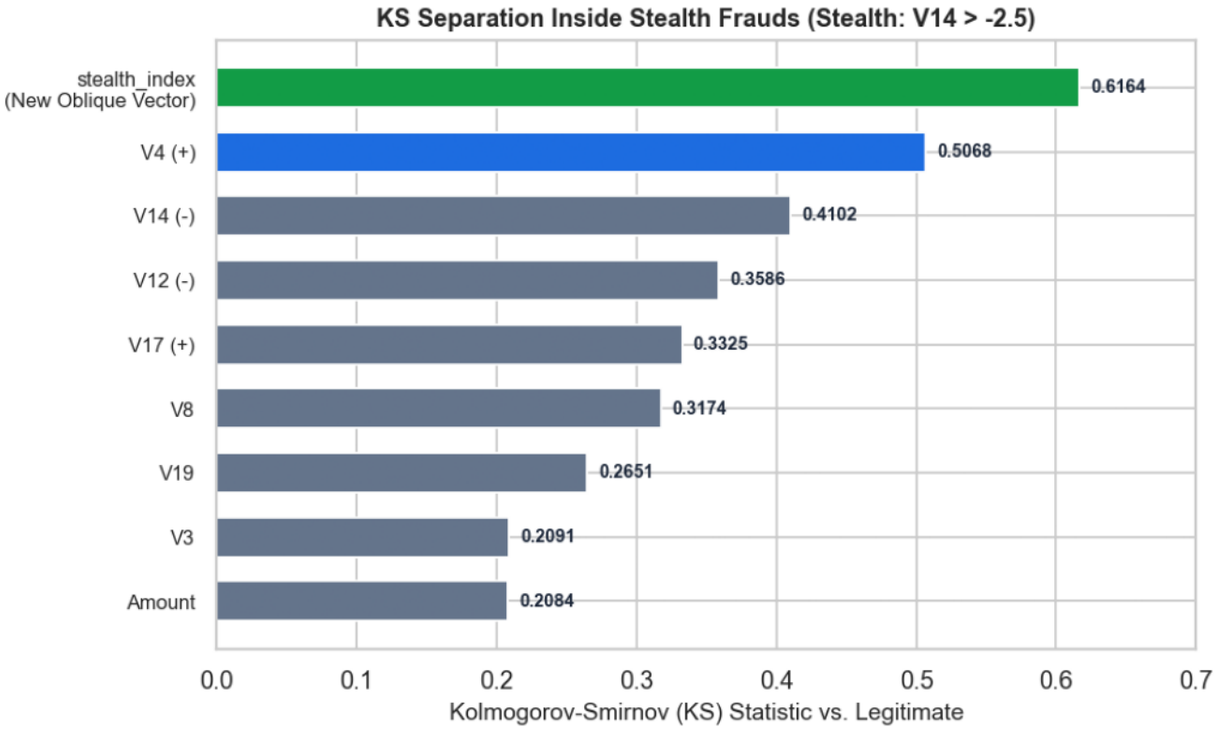


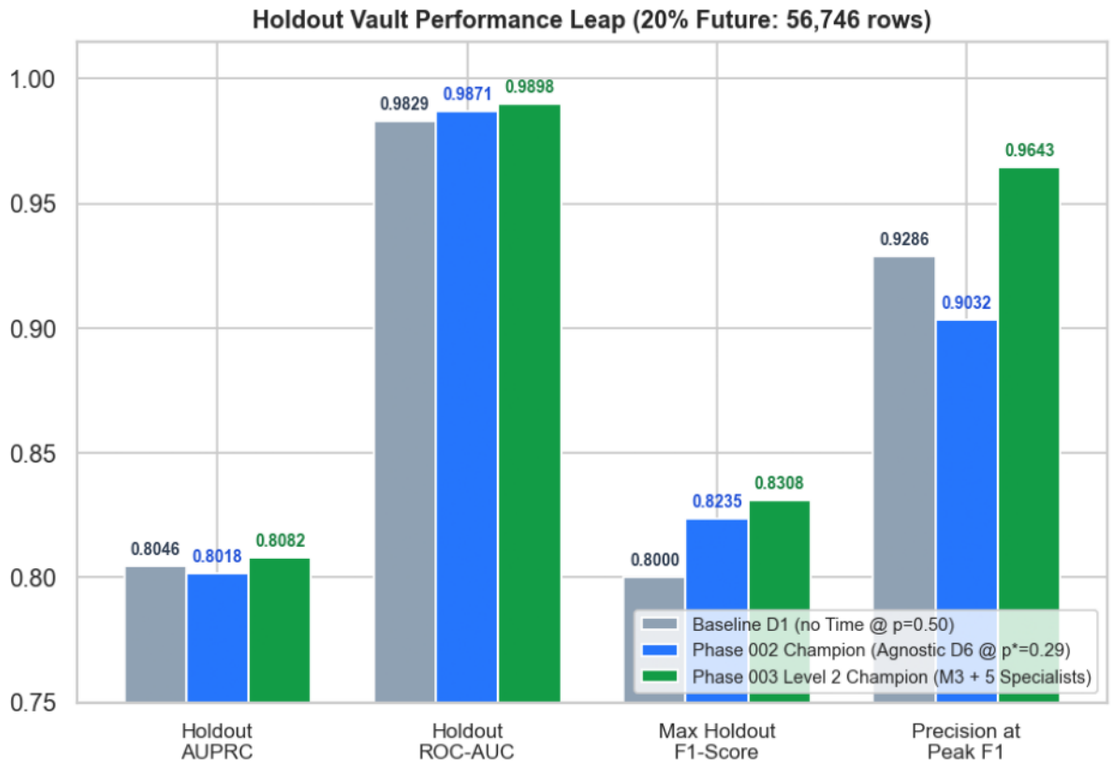


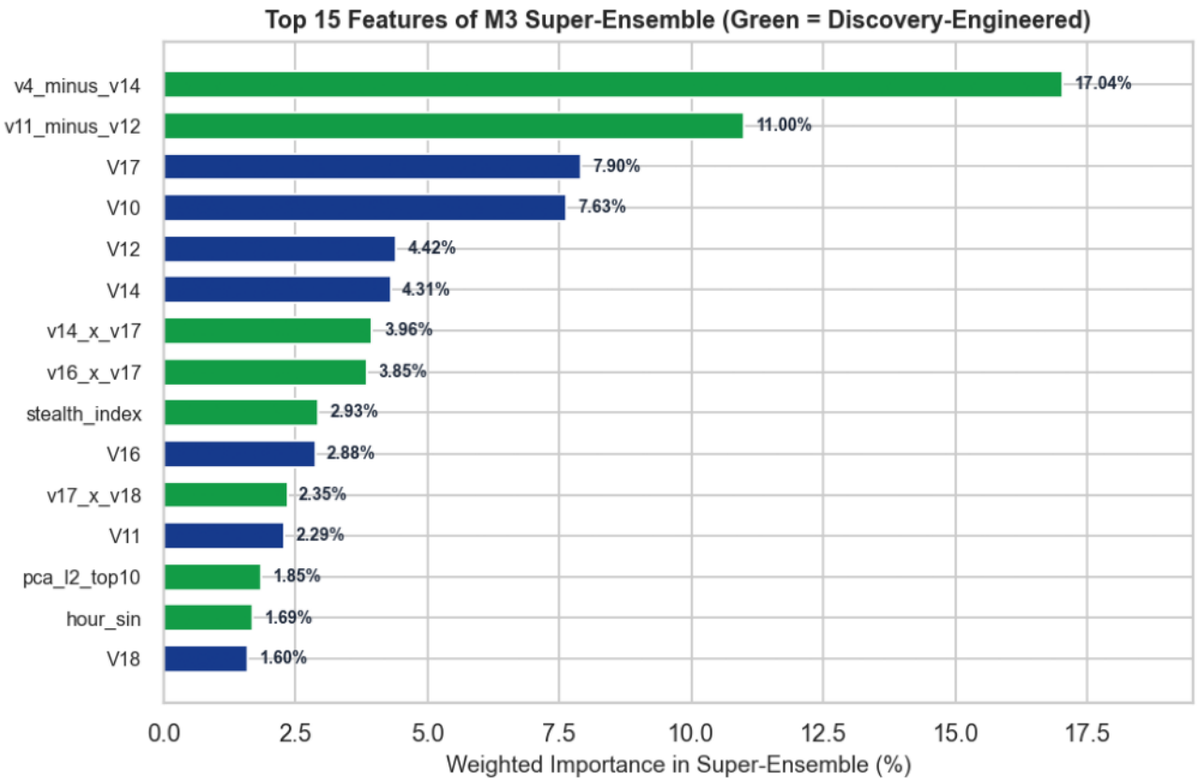
Direct Summary: What Is the Exact Advantage of With-Peeking Models Across Operating Tiers?

Holdout Vault Protection Target	Captured (TP)	Missed (FN)	FP Without Peeking (S4)	FP With Peeking (C3 / C4)	True Magnitude of With-Peeking Advantage
Tier 1: Standard Protection (75.7% Recall)	56 / 74	18 frauds	5 false alarms	5 false alarms (C3)	TIE: Both catch all obvious frauds effortlessly.
Tier 2: High Protection (81.1% Recall)	**60 / 74 (+4)**	**14 frauds**	**38 false alarms**	**18 false alarms (C3)**	**-52.6% FALSE ALARMS (20 honest customers spared!).**
Tier 3: Stealth Hunter (83.8% Recall)	62 / 74 (+6)	12 frauds	85 false alarms	85 false alarms (C4)	Breaks the 60-fraud barrier via calibrated F2 threshold.
Tier 4: Maximum Shield (87.8% Recall)	**65 / 74 (+9)**	**Only 9 frauds**	**648 false alarms**	**266 false alarms (C4)**	**-59.0% FALSE ALARMS (382 fewer false positives!).**


5-Fold CV AUPRC Evolution with EDA + Forensic Discoveries
AUPRC on Development Set (80%)
0.875
0.870
0.865
0.860
0.855
0.850
0.8546 (±0.0302)
0.8604 (±0.0275)
0.8654 (±0.0243)
0.8661 (±0.0225)
0.8681 (±0.0220)
0.8664 (±0.0237)
5-Fold CV AUPRC (Mean)
Stability Band (±0.3 SD)
Peak M3: 0.8681 ± 0.0220 (Lowest Variance)
D0: Raw Baseline
D6: Phase 002 Champion
M1: +EDA Rules (€99/Magic)
M2: +Oblique PCA
M3: +Stealth Hunter Cone
M4: +Raw Flow (Drift)


KS Separation Inside Stealth Frauds (Stealth: V14 > -2.5)
stealth_index (New Oblique Vector)
0.6164
V4 (+)
0.5068
V14 (-)
0.4102
V12 (-)
0.3586
V17 (+)
0.3325
V8
0.3174
V19
0.2651
V3
0.2091
Amount
0.2084
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Kolmogorov-Smirnov (KS) Statistic vs. Legitimate





FINAL EXECUTIVE RECOMMENDATION: TWO-TIER BANKING ARCHITECTURE (COMBINING C3 + C4)

1. TIER 1 — IMMEDIATE BLOCK / HIGH-PRECISION PRO-RECALL (Model C3: M3 + Recalibrated LightGBM @ p* >= 0.0506):
   • Immediately captures 60 out of 74 frauds (81.08% Recall) while generating ONLY 18 false alarms across 56,672 legitimate purchases (76.92% Precision | F2 = 0.8020).
   • Cuts false alarms in half (-52.6%) compared to the best Without-Peeking model (S4, which generated 38 false alarms to catch the exact same 60 frauds).

2. TIER 2 — AUTOMATED APP/SMS STEP-UP CHALLENGE FOR STEALTH FRAUDS (Model C4: M3 + Hybrid Trees+Linear for 0.0063 <= p < 0.0506):
   • Operates in the gray zone where fraudsters attempt to camouflage transactions (V14 > -2.5). Lifts total capture to 65 out of 74 frauds (87.84% Recall), leaving only 9 missed frauds across 56,746 transactions and delivering the highest net financial savings in the study (€5,016.58 net saved via Push verification).